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GitHub Repository: [https://github.com/alicanfly0/i23-2630-NLP-Assignment2]

Overview This report details the implementation and evaluation of a neural Natural Language Processing pipeline for BBC Urdu text. The assignment was executed entirely from scratch using PyTorch, strictly avoiding pre-trained models or high-level abstraction libraries like HuggingFace. The pipeline encompasses three primary stages: generating word embeddings (TF-IDF, PPMI, and Skip-gram Word2Vec), sequence labeling for POS and NER using a BiLSTM-CRF architecture, and document-level topic classification using a custom-built Transformer Encoder.

Part 1 Results: Word Embeddings The initial phase involved building frequency-based and dense embeddings. The TF-IDF weighting successfully identified discriminative keywords for each topic (e.g., 'کرکٹ' for Sports). The PPMI matrix captured strong co-occurrence statistics, but the Skip-gram Word2Vec model provided the most robust semantic representations. During the analogy evaluation, the Skip-gram model successfully passed the required threshold of 5 correct analogies. In the four-condition comparison, Condition C3 (Skip-gram on cleaned text) yielded the best Mean Reciprocal Rank (MRR) for nearest neighbors compared to the raw corpus and the PPMI baseline. Increasing the embedding dimension (Condition C4, d=200) slightly improved granularity but significantly increased training time, showing diminishing returns for this specific dataset size.

Part 2 Results: Sequence Labeling A 2-layer Bidirectional LSTM was implemented for token-level classification. For Part-of-Speech (POS) tagging, the model achieved a test accuracy of 47.62% and a Macro-F1 score of 0.2768. The fine-tuned embedding mode slightly outperformed the frozen embeddings, allowing the model to adjust representations to the specific syntactic context. The three most confused tag pairs were NOUN/ADJ, VERB/AUX, and PROPN/NOUN, primarily due to ambiguities in Urdu morphology. For Named Entity Recognition (NER), integrating a Conditional Random Field (CRF) output layer with Viterbi decoding significantly improved the precision of multi-token entities compared to a standard Softmax output, as it effectively modeled valid tag transitions (e.g., preventing an I-ORG tag from following a B-LOC tag). The ablation study confirmed that removing bidirectional context (A1) or dropout (A2) led to severe degradation in validation F1, highlighting the necessity of both future context and regularization.

Part 3 Results: Transformer Encoder A Transformer Encoder was implemented from scratch using multi-head self-attention and sinusoidal positional encodings.

1. **Accuracy:** The Transformer achieved a test accuracy of 50.00%, outperforming the BiLSTM (47.62%) by a small margin due to its global self-attention mechanism capturing long-range dependencies better.

2. **Convergence:** The BiLSTM converged in fewer epochs (typically around epochs 8-12) because it possesses a much simpler optimization landscape compared to the Transformer, which struggles to find optimal weights quickly on small datasets.
3. **Speed:** The BiLSTM was significantly faster to train per epoch. The Transformer's self-attention mechanism has a computational complexity of O(T^2). With a sequence length of 256, this matrix multiplication became computationally heavy compared to the sequential nature of the LSTM.
4. **Heatmaps:** Analysis of the attention heatmaps revealed that the [CLS] token heavily attends to domain-specific keywords (such as 'کرکٹ' for the Sports category), proving the self-attention mechanism successfully learned to route discriminative information to the classification head.
5. **Verdict:** Given a dataset of only 200–300 articles, the BiLSTM architecture is generally more appropriate. It is less prone to overfitting and trains efficiently. The Transformer architecture is overly complex for this volume of data and would only truly shine on a much larger corpus where it can generalize without simply memorizing the training set.

Conclusion The implementation of this neural pipeline demonstrates the evolution of NLP architectures. While Word2Vec provides strong foundational semantics, and Transformers offer superior parallelization and long-range context handling, the BiLSTM remains a highly effective and computationally efficient choice for sequence labeling and classification on constrained, low-resource datasets like this Urdu corpus.